

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · Avijeet Singh — Vortex AI · doi:10.5281/zenodo.20706893 · claims pinned to emem 64cfae5, re-verified live 29 Sep 2026



Hand over the address, not the sentence.

emem is **Earth memory for AI agents**. Every answer about a place — a pixel value, a field, a field through time — comes back as a **name derived from its own signed bytes**. Agents pass the name. Any receiver re-hashes and verifies it offline, **without trusting the sender, its model, or the server**.

WHAT AN AGENT GETS, TODAY

any lat/lng on Earth → signed facts in one call (Keylong: 4 layers in 0.2 s)

- 118** measurements from 46 sources — Sentinel-2, Copernicus DEM, JRC surface water, weather and more
- 10 m** signed rasters and cubes: the model gets a field, not a sentence about one
- 0.13 ms** to verify a receipt offline, on your own machine
- 114** MCP tools (18 core) + REST, one Rust binary

WHAT CHANGES INSIDE THE AGENT LOOP

- Context resets** checkpoint tokens, not paraphrased numbers; the next session resolves and re-verifies them [long-horizon-memory](#)
- Handoff** pass one 38-char bundle; the receiver checks every fact without trusting you [multi-agent-handoff](#)
- Before it speaks** “0.74” against a signed 28.0 → **drift: wrong**; a draft saying 35 °C → **denied** [referential-drift](#) · [verify-before-publish](#)

WHERE IT SITS IN YOUR GEOAI STACK

Agents	any model, any vendor — over MCP or REST
emem	identity · signed receipts · memory · append-only log
Sources	Sentinel-2 L2A, Copernicus DEM COGs, JRC water, embeddings — read upstream, never altered

Open source. Apache-2.0. Run it tonight.

MCP · Python · TypeScript · Docker · LlamaIndex · LangMem — commands at the bottom of this poster

1 · THE INVARIANT

The bytes are their own name.

```
{ kind: "primary",
  cell: "defi.zb493.xuqA.zcb5f", // Bengaluru
  band: "copdem30m.elevation_mean",
  tslot: 0, // valid time
  value: 918.0, unit: "m",
  sources: [{ scheme: "open_meteo" }],
  derivation: { fn_key: "open_meteo_copdem90m@1" },
  signer: ed25519:777et3yi...,
  signed_at: "2026-05-28T19:54:32Z" // txn time }
```

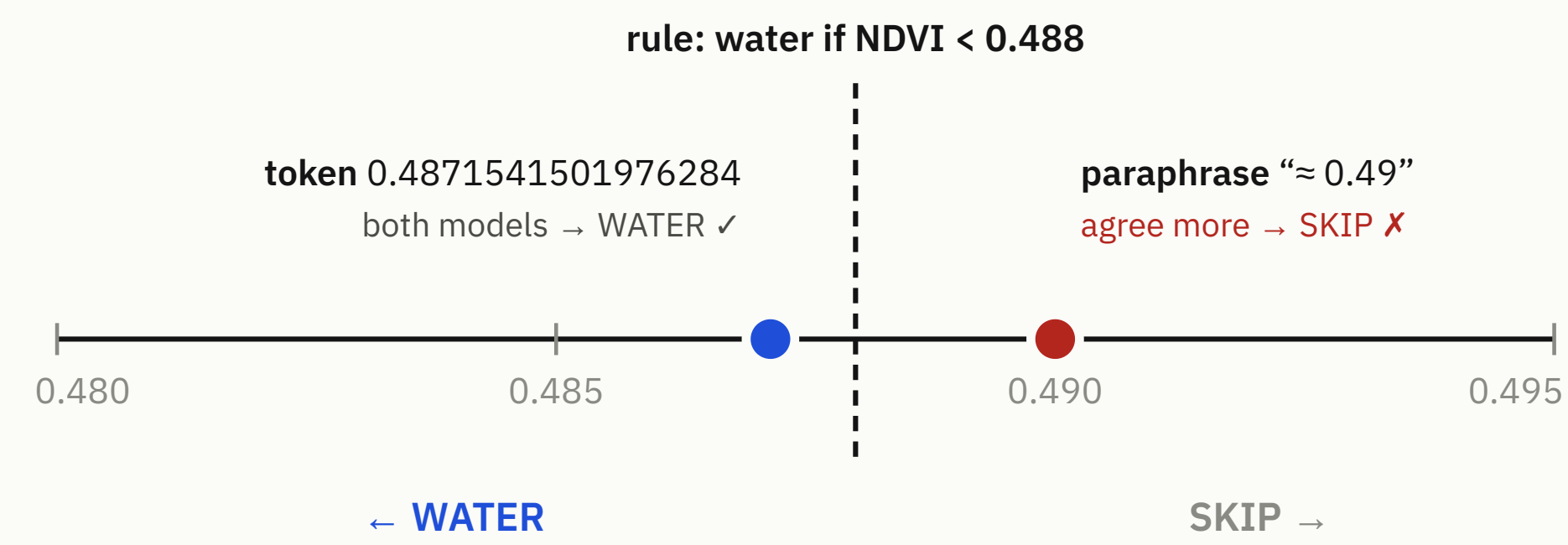
ENCODE emem-CBOR, 509 bytes
HASH BLAKE3-256 → base32
NAME yqbo1geoycqkvj3zkxukb4bjw4odhpwvfzqo3fbgwf4spk45za1a

- Same bytes → same name. Stock BLAKE3, no emem code, no registry.
- 918.0 → 918.1 gives a different name (vines361L).
- Cited under the wrong cell → HTTP 409. Never a silent wrong answer.

Deterministic CBOR in declaration order, NaN/-0 canonical (not RFC 8949 key-sorted), signer and signed_at are hashed, so a CID names one signed attestation; the ed25519 signature covers a Merkle root of fact CIDs.

4 · THE FINDING

A paraphrase made two models agree — and both chose wrong.



Pre-registered: after a shared summary	correct · agree
full context (control)	72/72 · 36/36
compaction, free reasoning	20/72 · 15/36
compaction under pressure	0/72 · 3/36, p = 0.035

Agreement between agents is not evidence of correctness. It is what a shared lossy summary produces.

One real Lahaul NDVI fact; Gemma-4-12B and Qwen2.5-7B on one host; Fisher one-sided; six scorer bugs fixed before the verdict.

Use it tonight. It’s open source.

Every command below was run from a clean install on 29 Sep 2026. Apache-2.0 — fork it, self-host it, break it, tell us.

Your agent · any MCP client

```
claude mcp add \
--transport http \
emem https://emem.dev/mcp

# then ask it about
# any place on Earth
```

18 core tools; 114 at /mcp/full.

Python · ememdev 2.4.2

```
# pip install "ememdev[signing]"
from ememdev import Client
from ememdev.verify import \
    verify_receipt_offline

em = Client()
c = em.locate("Keylong")["cell64"]
r = em.recall(c, bands=["indices.ndvi"])
verify_receipt_offline(
    r["receipt"]).ok # True
```

TypeScript: npm i @vortexai/emem

Your own node · Docker

```
docker run -p 5051:5051 \
ghcr.io/vortex-ai/emem:latest

# your key, your receipts;
# air-gapped image too:
# ghcr.io/vortex-ai/emem-airgap
```

One static Rust binary. amd64 + arm64.

Trust nothing · stock BLAKE3

```
cid = "yqbo1geoycqkvj3z..."
body = GET("emem.dev/v1/facts/"+cid,
accept="application/cbor")
b32(blake3(body)) == cid

# True: the bytes are
# their own name
```

Full script behind the first QR.

2 · THE RECEIVER

Resolve. Don’t trust.

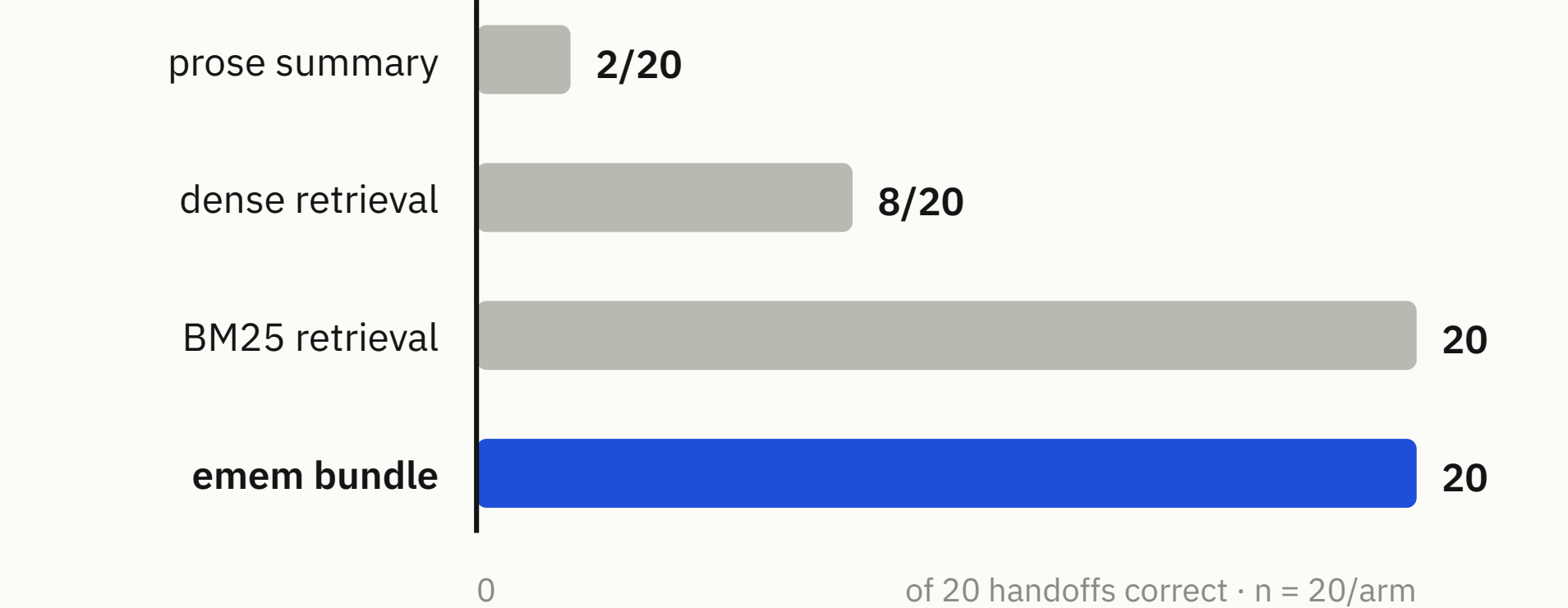
- Resolve the token → the exact bytes
GET /v1/facts/<cid>
- Re-hash them yourself. Does the name match?
- Bind the place: the token’s cell must equal the cell inside the signed fact.
- Verify the receipt offline: ed25519 over a domain-separated, length-prefixed preimage.
- Check history: RFC 6962-style log, 2.54 M entries, co-signed by an independent operator (geo.qa).

CROSSES	DOES NOT NEED TO
84 characters — or one emem:bundle: handle, 38 chars for up to 256 facts.	the sender’s prompt, context, model, database, or honesty.

A proof-less receipt signs an explicit ABSENT marker, so stripping a proof is detectable. An independent audit rejected 5/5 tampers, including a version downgrade.

5 · HANDOFF BETWEEN AGENTS

What Agent B gets right, by what A handed over.



Long run: 599 turns, n = 1,624/arm	within 1% · citable
emem bundle	100% · 100%
BM25 retrieval	35% · 0%

BM25 ties on short handoffs — we say it plainly. What it cannot hand the next agent is a citation that re-checks; over long runs it drifts.

multiagent_v2 and long-run harness, Gemma ↔ Qwen, both directions. Dense top-5 retrieval: median location error 252 m.

3 · FOUND IN PRODUCTION, NOT STAGED

One place. Two answers. Both checkable.

Token signed 28 May 2026

918.0 m

Bengaluru · source open_meteo, Copernicus DEM 90 m
emem:fact:defi.zb493.xuqA.zcb5f:yqbo1geo...
re-hash 29 Sep: MATCH

↓ the upstream provider changed

Live answer, signed 28 Sep 2026

915.07 m

same cell, same band · source Copernicus DEM 30 m COG
emem:fact:defi.zb493.xuqA.zcb5f:jzxzmvom...
re-hash 29 Sep: MATCH

The sentence “*elevation ≈ 918 m*” cannot tell you the answer moved, or why. The token still returns 918.0, with the dataset and recipe that produced it.

Spotted by an external auditor; re-verified by us with the blake3 and cbor2 reference libraries. Corrections are new facts with new names; nothing is overwritten.

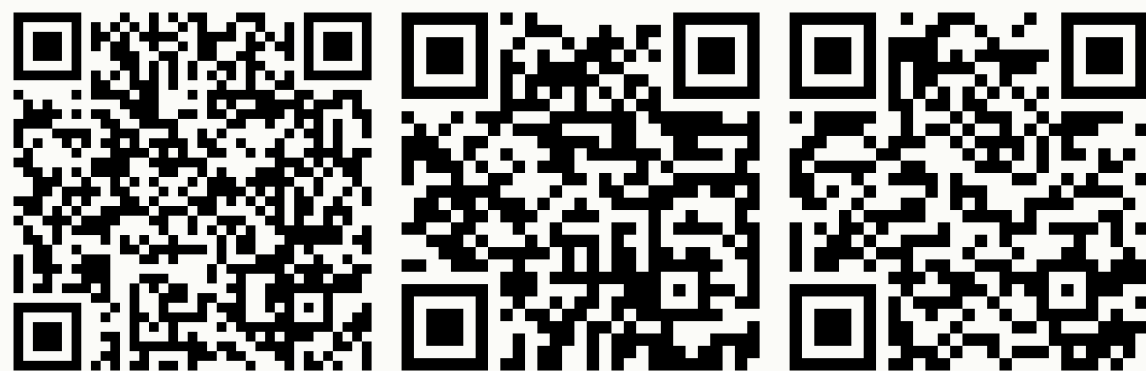
6 · BOUNDARY OF THE CLAIM

What holds. What doesn’t.

CHECKABLE	NOT PROVED
✓byte integrity of a cited fact or field	✗that the observation is true
✓where and when it claims to be	✗sensor or model accuracy
✓who signed it, and when	✗two sensors saw one thing
✓history not rewritten	✗a sentence cites the right band
✓“nothing here” — signed absence	✗anything in orbit (edge signing is simulated: 0.17% of a 40 ms frame)
✓disagreement kept and scored	

Our own scorecard	
Citation survives compaction; paraphrase does not	supported
Beats dense retrieval	supported
Beats plain context · beats BM25	refuted
A single token saves context (bundles do)	refuted
Beats peer agent-memory products	not tested

Embeddings (live: TESSERA 128-D; earlier Clay v1.5, Prithvi-EO-2.0, Galileo) are signed as model1_output facts — a learned representation, not a measurement. Open: formal proofs, full conformance vectors, hardware attestation.



Numbers from emem docs (benchmarks, how-emem-compares, paper-section-statistics): single node, open models, no independent replication yet. 216 agents and 8,427 signed notes in the public collaboration log. Browser verifier: emem.dev/verify.